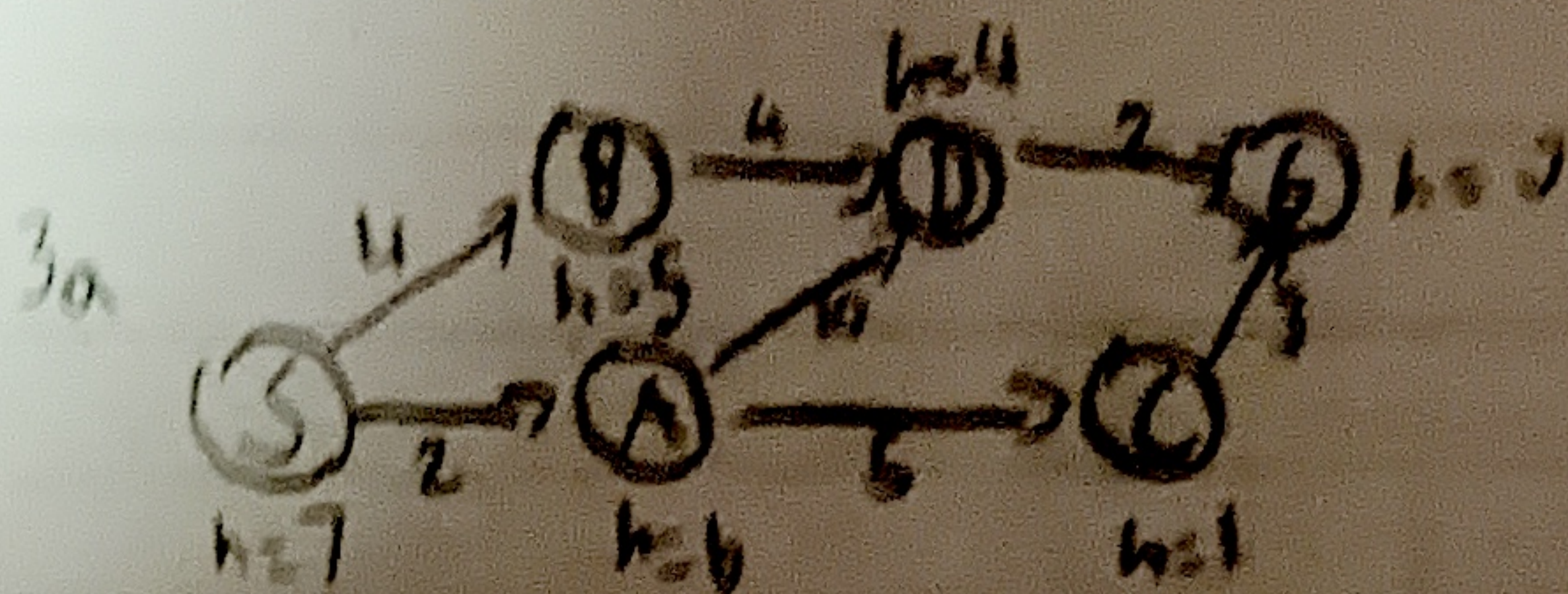


Assignment 2:

1. If we are looking at the graphs, we can see that the path for both BFS and DFS is the same: $A \rightarrow B \rightarrow C \rightarrow E$. However, the visiting order is different. In BFS, after B, it visits C, D, then E. In DFS, it visits D, F, C, then E. BFS travels level by level while DFS discovered D before C and went to F because of stack (LIFO). BFS ended up getting to the goal faster since it didn't even look at F.

For the grid, DFS actually took less time to travel to its goal compared to BFS by going down to E directly to the shorter path.

2. Although limits 6 and 10 were able to reach the goal fine, if I used limits 5 and that's when it couldn't. The path is similar to DFS, except that with limit 5 when the node couldn't be reached, it took the other path, which it also didn't reach.



36. F: S(7)
E:

F: A(9), B(2)
E: S(7)

F: B(2), C(8), D(16)
E: S(7), A(8)

F: B(2), D(16), G(10)
E: S(7), A(8), C(8)

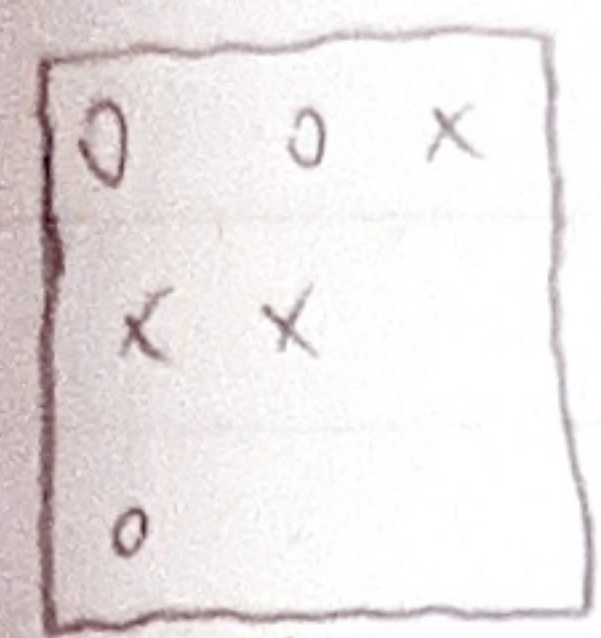
F: B(10), D(12)
E: S(7), A(8), C(8), B(2)

F:
E: S(7), A(8), C(8), B(2), G(10)

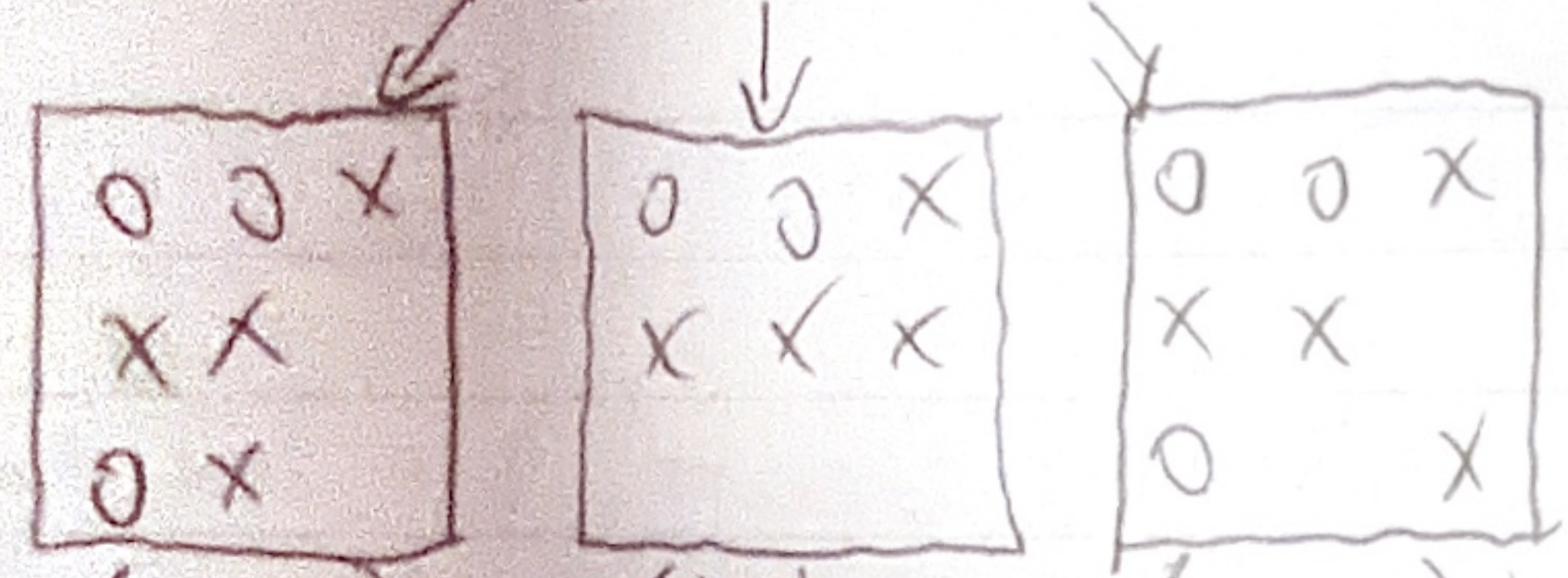
Path Explored: S → A → C → E
Cost: 10

a. a.

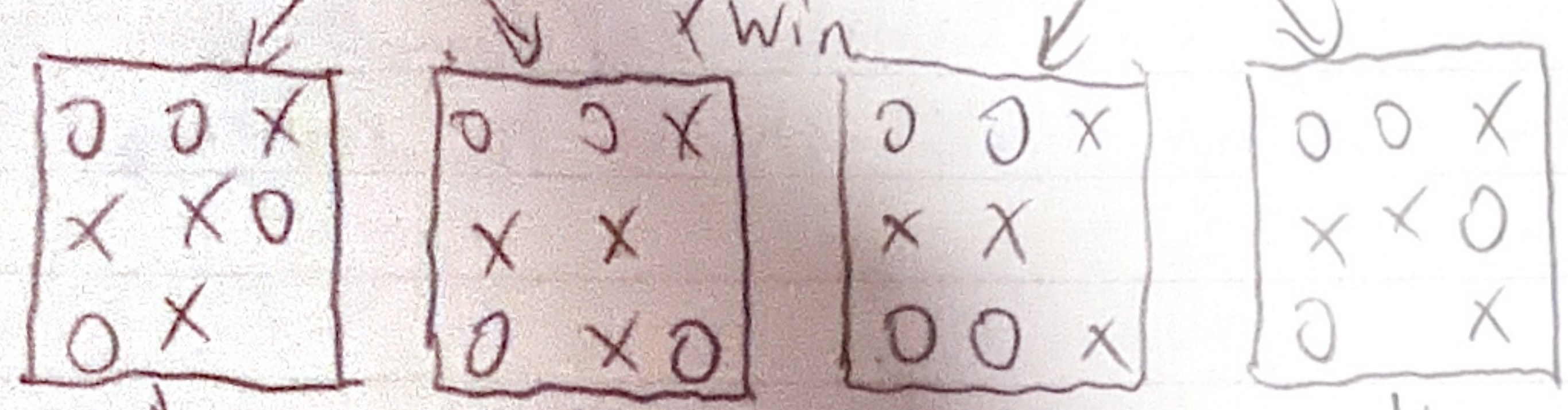
1
base



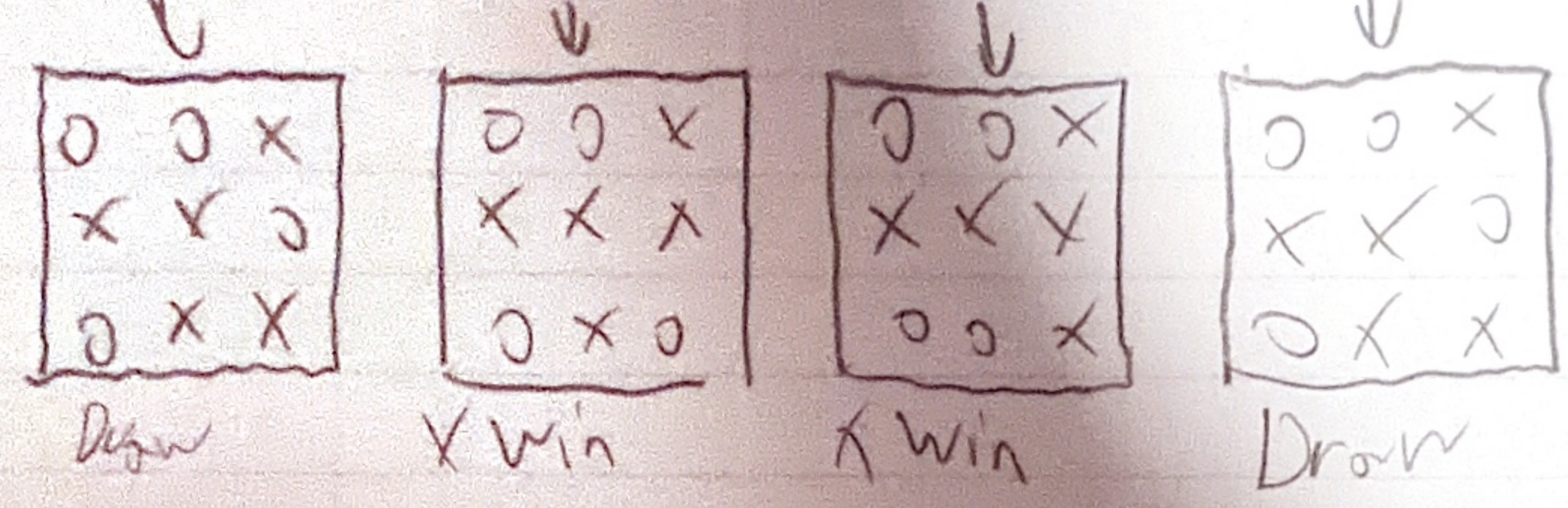
2
x turn



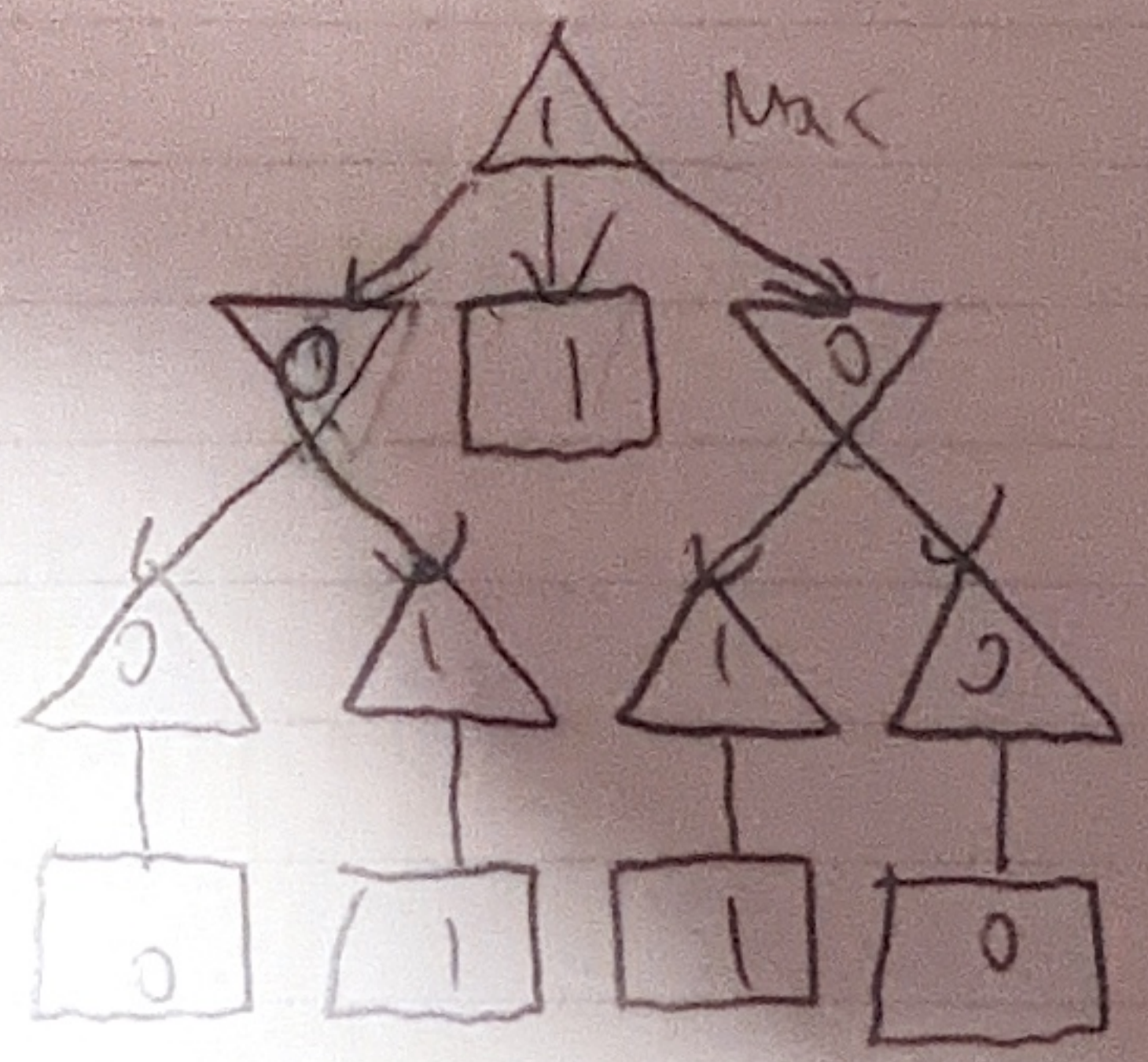
3
o turn



4
x turn



- b. 4 layers
- c. 5 terminals
- d.



5.



Nodes 15 and 7 can be pruned because 2 is less than 3, which 3 is already checked, so min is established and doesn't need to check the rest of its neighbors. It still checks the rightmost tree however.